Homework Solutions: Lectures 19–21

The z-Transform: Definition, Inversion, Properties, and System Analysis

CEC 315 – Signals and Systems

Rogelio Gracia Otalvaro

Spring 2026

Problem 1: Computing z-Transforms and ROCs

Part (a): $x_1[n] = 5\,(0.7)^n\,u[n]$

Using the pair $a^n u[n] \mathcal{Z} \frac{1}{1-az^{-1}}$ with $a = 0.7$:

$$\boxed{X_1(z) = \frac{5}{1 - 0.7z^{-1}}\,, \qquad |z| > 0.7}$$

Pole at $z = 0.7$. Check: $X_1(1) = 5/(1 - 0.7) = 5/0.3 = 50/3$.

Part (b): $x_2[n] = -(4)^n\,u[-n-1]$

Using the pair $-a^n u[-n-1] \mathcal{Z} \frac{1}{1-az^{-1}}$ with $a = 4$, ROC: $|z| < |a|$:

$$\boxed{X_2(z) = \frac{1}{1 - 4z^{-1}}\,, \qquad |z| < 4}$$

Pole at $z = 4$.

Check: $X_2(1) = 1/(1 - 4) = -1/3$. Direct: $\sum_{n=-\infty}^{-1} -(4)^n = -\sum_{m=1}^{\infty} 4^{-m} = -\frac{1/4}{1-1/4} = -1/3$. ✓

Part (c): $x_3[n] = 3\,\delta[n] - 2\,\delta[n-1] + \delta[n-4]$

Using $\delta[n-k] \mathcal{Z} z^{-k}$:

$$\boxed{X_3(z) = 3 - 2z^{-1} + z^{-4}\,, \qquad |z| > 0}$$

This is a finite-duration sequence (polynomial in z^{-1}). ROC is all $z \neq 0$.

Check: $X_3(1) = 3 - 2 + 1 = 2 = 3 - 2 + 0 + 0 + 1$. ✓

Part (d): $x_4[n] = n\,(0.5)^n\,u[n]$

Using the pair $n\,a^n u[n] \mathcal{Z} \frac{az^{-1}}{(1-az^{-1})^2}$ with $a = 0.5$:

$$\boxed{X_4(z) = \frac{0.5\,z^{-1}}{(1-0.5z^{-1})^2}\,, \qquad |z| > 0.5}$$

Double pole at $z = 0.5$, zero at $z = 0$.

Check: $X_4(1) = \frac{0.5}{(0.5)^2} = \frac{0.5}{0.25} = 2$. Direct: $\sum_{n=0}^{\infty} n(0.5)^n = \frac{0.5}{(1-0.5)^2} = 2$. ✓

Part (e): $x_5[n] = 2\,(0.3)^n\,u[n] + 4\,(0.9)^n\,u[n]$

Step 1: Transform each term:

$$\frac{2}{1-0.3z^{-1}}\,,\ |z| > 0.3 \qquad \text{and} \qquad \frac{4}{1-0.9z^{-1}}\,,\ |z| > 0.9$$

Step 2: Common denominator:

$$X_5(z) = \frac{2(1-0.9z^{-1}) + 4(1-0.3z^{-1})}{(1-0.3z^{-1})(1-0.9z^{-1})} = \frac{6-3.0z^{-1}}{(1-0.3z^{-1})(1-0.9z^{-1})}$$

ROC: $|z| > 0.3 \cap |z| > 0.9 = |z| > 0.9$.

$$\boxed{X_5(z) = \frac{6-3z^{-1}}{(1-0.3z^{-1})(1-0.9z^{-1})}\,, \qquad |z| > 0.9}$$

Poles: $z = 0.3$ and $z = 0.9$. Zero: $6 - 3z^{-1} = 0 \Rightarrow z = 0.5$.

Check: $X_5(1) = \frac{6-3}{(0.7)(0.1)} = \frac{3}{0.07} = 300/7 \approx 42.86$. Direct: $\frac{2}{0.7} + \frac{4}{0.1} = \frac{20}{7} + 40 = \frac{300}{7}$. ✓

Part (f): $x_6[n] = (0.6)^n\,u[n] - 3\,(2)^n\,u[-n-1]$

Step 1:

$$(0.6)^n u[n] \ \mathcal{Z}\ \frac{1}{1-0.6z^{-1}}\,, \quad |z| > 0.6$$

$$-3(2)^n u[-n-1] = -3 \cdot \big[-(2)^n u[-n-1]\big] \cdot (-1) \ \mathcal{Z}\ \frac{3}{1-2z^{-1}}\,, \quad |z| < 2$$

Wait, let us be careful. We have $-3 \cdot (2)^n u[-n-1]$. Write this as $(-3) \cdot \big[-(2)^n u[-n-1]\big] \cdot (-1)$. Actually: $-3(2)^n u[-n-1] = 3 \cdot \big[-(2)^n u[-n-1]\big]$. Since $-(2)^n u[-n-1] \mathcal{Z} \frac{1}{1-2z^{-1}}$ with ROC $|z| < 2$, we get:

$$3 \cdot \frac{1}{1-2z^{-1}} = \frac{3}{1-2z^{-1}}\,, \quad |z| < 2$$

Step 2:

$$X_6(z) = \frac{1}{1 - 0.6z^{-1}} + \frac{3}{1 - 2z^{-1}}$$

ROC: $|z| > 0.6 \ \cap \ |z| < 2 = \{z : 0.6 < |z| < 2\}$.

Since the unit circle ($|z| = 1$) satisfies $0.6 < 1 < 2$, the **DTFT exists**. ✓

$$\boxed{X_6(z) = \frac{1}{1 - 0.6z^{-1}} + \frac{3}{1 - 2z^{-1}}, \qquad 0.6 < |z| < 2}$$

Part (g): Sanity Checks at $z = 1$

All checks were performed inline above. The rule $X(1) = \sum_n x[n]$ holds because $z^{-n}\big|_{z=1} = 1$ for all n.

Problem 2: Inverse z-Transform via Partial Fractions

Part (a): Distinct Real Poles, Causal

Given: $X(z) = \dfrac{1+3z^{-1}}{(1-0.2z^{-1})(1-0.6z^{-1})}$, ROC: $|z| > 0.6$.

Step 1: Partial fractions.

$$\frac{1+3z^{-1}}{(1-0.2z^{-1})(1-0.6z^{-1})} = \frac{A}{1-0.2z^{-1}} + \frac{B}{1-0.6z^{-1}}$$

Set $z^{-1} = 1/0.2 = 5$: $1 + 15 = A(1-3) \Rightarrow 16 = -2A \Rightarrow A = -8$.

Set $z^{-1} = 1/0.6 = 5/3$: $1 + 5 = B(1 - 1/3) \Rightarrow 6 = (2/3)B \Rightarrow B = 9$.

Step 2: ROC is $|z| > 0.6$. Both poles ($|0.2|, |0.6| < 0.6$...). Wait: $|0.2| = 0.2 < 0.6$ and $|0.6| = 0.6$. The ROC $|z| > 0.6$ is outside both pole circles, so both terms are right-sided.

Step 3:

$$\boxed{x[n] = \big[-8\,(0.2)^n + 9\,(0.6)^n\big]\,u[n]}$$

Check: $x[0] = -8 + 9 = 1$. Initial-value theorem: $\lim_{z\to\infty} X(z) = 1/1 = 1$. ✓

Part (b): Two-Sided Signal

Given: $X(z) = \dfrac{4}{(1-0.5z^{-1})(1-3z^{-1})}$, ROC: $0.5 < |z| < 3$.

Step 1: Partial fractions.

$$\frac{4}{(1-0.5z^{-1})(1-3z^{-1})} = \frac{A}{1-0.5z^{-1}} + \frac{B}{1-3z^{-1}}$$

Set $z^{-1} = 2$: $4 = A(1-6) = -5A \Rightarrow A = -4/5$.

Set $z^{-1} = 1/3$: $4 = B(1 - 1/6) = (5/6)B \Rightarrow B = 24/5$.

Step 2: Assign directions from the annular ROC $0.5 < |z| < 3$:

Pole at $z = 0.5$ ($|z| = 0.5$): the ROC is **outside** this pole $\Rightarrow$ right-sided.

Pole at $z = 3$ ($|z| = 3$): the ROC is **inside** this pole $\Rightarrow$ left-sided.

Step 3:

$$\frac{-4/5}{1-0.5z^{-1}}\ \text{(right-sided)} \longrightarrow -\frac{4}{5}\,(0.5)^n\,u[n]$$

$$\frac{24/5}{1-3z^{-1}}\ \text{(left-sided)} \longrightarrow -\frac{24}{5}\,(3)^n\,u[-n-1]$$

$$\boxed{x[n] = -\frac{4}{5}\,(0.5)^n\,u[n] - \frac{24}{5}\,(3)^n\,u[-n-1]}$$

Key Point

The left-sided inversion uses the pair $\frac{B}{1-az^{-1}}$ with ROC $|z| < |a| \longleftrightarrow -B\,a^n\,u[-n-1]$. The negative sign is part of the pair itself, so $B = 24/5$ gives $-(24/5)(3)^n u[-n-1]$.

Part (c): Repeated Poles

Given: $X(z) = \dfrac{z^{-1}}{(1-0.5z^{-1})^2}$, ROC: $|z| > 0.5$.

Using the pair $n\,a^n\,u[n] \mathcal{Z} \frac{az^{-1}}{(1-az^{-1})^2}$ with $a = 0.5$:

We can write $X(z) = \frac{z^{-1}}{(1-0.5z^{-1})^2} = \frac{1}{0.5} \cdot \frac{0.5z^{-1}}{(1-0.5z^{-1})^2} = 2 \cdot \frac{0.5z^{-1}}{(1-0.5z^{-1})^2}$.

$$\boxed{x[n] = 2\,n\,(0.5)^n\,u[n]}$$

Check: $x[0] = 0$, $x[1] = 2(1)(0.5) = 1$, $x[2] = 2(2)(0.25) = 1$.

Initial-value theorem: $\lim_{z\to\infty} X(z) = 0$. ✓ ($x[0] = 0$, consistent).

Part (d): Complex Conjugate Poles

Given: $X(z) = \dfrac{1 - 0.9\cos(0.3\pi)\,z^{-1}}{1 - 2(0.9)\cos(0.3\pi)\,z^{-1} + 0.81\,z^{-2}}$, ROC: $|z| > 0.9$.

Comparing with the table pair for $r^n \cos(\omega_0 n)\,u[n]$:

$$\frac{1 - r\cos\omega_0\,z^{-1}}{1 - 2r\cos\omega_0\,z^{-1} + r^2 z^{-2}}$$

we identify: $r = 0.9$, $\omega_0 = 0.3\pi$, $r^2 = 0.81$. ✓

Poles at $z = 0.9\,e^{\pm j0.3\pi}$, both with $|z| = 0.9 < 1$.

$$\boxed{x[n] = (0.9)^n \cos(0.3\pi\,n)\,u[n]}$$

This is a damped discrete-time sinusoid: the $(0.9)^n$ envelope decays because $r < 1$, and the oscillation frequency is $\omega_0 = 0.3\pi$ rad/sample.

Problem 3: z-Transform Properties

Part (a): Time Shifting

Given: $(0.8)^n u[n] \mathcal{Z} \frac{1}{1-0.8z^{-1}}$, $|z| > 0.8$.

By the time-shift property, $x[n-2] \mathcal{Z} z^{-2} X(z)$:

$$\boxed{Y(z) = \frac{z^{-2}}{1-0.8z^{-1}}, \qquad |z| > 0.8}$$

The factor z^{-2} adds **two zeros at** $z = 0$. In the z-form: $Y(z) = \frac{z^{-2}}{1-0.8z^{-1}} = \frac{1}{z^2-0.8z} = \frac{1}{z(z-0.8)}$. So there is one pole at $z = 0.8$, one pole at $z = 0$, and the two zeros from z^{-2} cancel one of the $z = 0$ poles... actually, rewriting: $Y(z) = z^{-2} \cdot \frac{z}{z-0.8} = \frac{z^{-1}}{z-0.8} = \frac{1}{z^2-0.8z}$. Zeros: $z = \infty$ (two of them, since numerator degree $<$ denominator degree by 2). In the z^{-1} form, z^{-2} contributes 2 zeros at $z = 0$.

Part (b): z-Domain Scaling

Given: $u[n] \mathcal{Z} \frac{1}{1-z^{-1}}$, $|z| > 1$.

Scaling property: $z_0^n x[n] \mathcal{Z} X(z/z_0)$. Set $z_0 = 0.5$:

$(0.5)^n u[n] \mathcal{Z} X(z/0.5) = \frac{1}{1-(z/0.5)^{-1}} = \frac{1}{1-0.5z^{-1}}$

ROC: $|z/0.5| > 1 \Rightarrow |z| > 0.5$.

$$\boxed{(0.5)^n u[n] \;\; \mathcal{Z} \;\; \frac{1}{1-0.5z^{-1}}, \quad |z| > 0.5}$$

This matches the standard pair with $a = 0.5$. ✓

Part (c): Convolution Property

(i) $H_1(z) = \frac{1}{1-0.3z^{-1}}$, $|z| > 0.3$. $\quad H_2(z) = \frac{1}{1-0.7z^{-1}}$, $|z| > 0.7$.

(ii) $H(z) = \frac{1}{(1-0.3z^{-1})(1-0.7z^{-1})}$, $|z| > 0.7$.

(iii) Partial fractions:

$$\frac{1}{(1-0.3z^{-1})(1-0.7z^{-1})} = \frac{A}{1-0.3z^{-1}} + \frac{B}{1-0.7z^{-1}}$$

Set $z^{-1} = 10/3$: $1 = A(1-7/3) = A(-4/3) \Rightarrow A = -3/4$.

Set $z^{-1} = 10/7$: $1 = B(1-3/7) = B(4/7) \Rightarrow B = 7/4$.

$$\boxed{h[n] = \left[-\frac{3}{4}(0.3)^n + \frac{7}{4}(0.7)^n\right] u[n]}$$

(iv) Verify by direct convolution:

$h[0] = (h_1 * h_2)[0] = h_1[0] \cdot h_2[0] = 1 \cdot 1 = 1.$

From formula: $h[0] = -3/4 + 7/4 = 4/4 = 1.$ ✓

$h[1] = h_1[0]h_2[1] + h_1[1]h_2[0] = 1 \cdot 0.7 + 0.3 \cdot 1 = 1.0.$

From formula: $h[1] = -3/4 \cdot 0.3 + 7/4 \cdot 0.7 = -0.225 + 1.225 = 1.0.$ ✓

Part (d): Initial-Value Theorem and Full Inversion

Given: $X(z) = \dfrac{3 - z^{-1}}{(1 - 0.4z^{-1})(1 + 0.5z^{-1})}$, $|z| > 0.5$.

(i) Initial value: $x[0] = \lim_{z\to\infty} X(z) = \frac{3-0}{(1-0)(1+0)} = 3.$

$$\boxed{x[0] = 3}$$

(ii) Partial fractions. Poles at $z = 0.4$ and $z = -0.5$.

$$\frac{3 - z^{-1}}{(1 - 0.4z^{-1})(1 + 0.5z^{-1})} = \frac{A}{1 - 0.4z^{-1}} + \frac{B}{1 + 0.5z^{-1}}$$

Set $z^{-1} = 1/0.4 = 2.5$: $3 - 2.5 = A(1 + 1.25) = 2.25A \Rightarrow A = 0.5/2.25 = 2/9.$

Set $z^{-1} = -1/0.5 = -2$: $3 + 2 = B(1 + 0.8) = 1.8B \Rightarrow B = 5/1.8 = 25/9.$

$$x[n] = \frac{2}{9}(0.4)^n\, u[n] + \frac{25}{9}(-0.5)^n\, u[n]$$

Check: $x[0] = 2/9 + 25/9 = 27/9 = 3.$ ✓

(iii) $x[1] = \frac{2}{9}(0.4) + \frac{25}{9}(-0.5) = \frac{0.8}{9} - \frac{12.5}{9} = \frac{-11.7}{9} = -1.3.$

$x[2] = \frac{2}{9}(0.16) + \frac{25}{9}(0.25) = \frac{0.32+6.25}{9} = \frac{6.57}{9} = 0.73.$

Problem 4: System Analysis with $H(z)$

System: $y[n] - 0.9\,y[n-1] + 0.18\,y[n-2] = x[n]$.

Part (a): System Function

Replace delays with z^{-k}:

$$(1 - 0.9z^{-1} + 0.18z^{-2})Y(z) = X(z)$$

Factor: $z^2 - 0.9z + 0.18 = 0 \Rightarrow z = \frac{0.9 \pm \sqrt{0.81-0.72}}{2} = \frac{0.9 \pm 0.3}{2}$.

Poles: $z = 0.6$ and $z = 0.3$. No finite zeros (numerator is 1).

$$\boxed{H(z) = \frac{1}{(1-0.6z^{-1})(1-0.3z^{-1})}, \qquad |z| > 0.6 \text{ (causal)}}$$

Part (b): Stability

Both poles satisfy $|0.6| < 1$ and $|0.3| < 1$. Since the system is causal with all poles strictly inside the unit circle, it is **BIBO stable**. ✓

Part (c): Impulse Response

$$\frac{1}{(1-0.6z^{-1})(1-0.3z^{-1})} = \frac{A}{1-0.6z^{-1}} + \frac{B}{1-0.3z^{-1}}$$

Set $z^{-1} = 5/3$: $1 = A(1-0.5) = 0.5A \Rightarrow A = 2$.

Set $z^{-1} = 10/3$: $1 = B(1-2) = -B \Rightarrow B = -1$.

$$\boxed{h[n] = \left[2(0.6)^n - (0.3)^n\right] u[n]}$$

Check: $h[0] = 2 - 1 = 1$ ✓ (coefficient of $x[n]$ in the ODE is 1).

Part (d): Output for Input $(0.5)^n u[n]$

(i) $X(z) = \frac{1}{1-0.5z^{-1}}$, $|z| > 0.5$.

$$Y(z) = \frac{1}{(1-0.6z^{-1})(1-0.3z^{-1})(1-0.5z^{-1})}, \quad |z| > 0.6$$

(ii) Partial fractions:

$$\frac{1}{(1-0.6z^{-1})(1-0.3z^{-1})(1-0.5z^{-1})} = \frac{A}{1-0.6z^{-1}} + \frac{B}{1-0.3z^{-1}} + \frac{C}{1-0.5z^{-1}}$$

Set $z^{-1} = 5/3$: $1 = A(1-0.5)(1-5/6) = A(0.5)(1/6) = A/12 \Rightarrow A = 12$.

Set $z^{-1} = 10/3$: $1 = B(1-2)(1-5/3) = B(-1)(-2/3) = 2B/3 \Rightarrow B = 3/2$.

Set $z^{-1} = 2$: $1 = C(1-1.2)(1-0.6) = C(-0.2)(0.4) = -0.08C \Rightarrow C = -12.5$.

Check: $A + B + C = 12 + 1.5 - 12.5 = 1$. $Y(\infty) = A + B + C = 1 = y[0]$. We need $y[0] = x[0] = 1$ from the ODE with zero ICs. ✓

$$\boxed{y[n] = \left[12\,(0.6)^n + 1.5\,(0.3)^n - 12.5\,(0.5)^n\right] u[n]}$$

(iii) Verify:

$y[0] = 12 + 1.5 - 12.5 = 1$. From ODE: $y[0] = 0.9 \cdot 0 - 0.18 \cdot 0 + x[0] = 1$. ✓

$y[1] = 12(0.6) + 1.5(0.3) - 12.5(0.5) = 7.2 + 0.45 - 6.25 = 1.4$.

From ODE: $y[1] = 0.9 \cdot y[0] - 0.18 \cdot 0 + x[1] = 0.9(1) + 0.5 = 1.4$. ✓

Part (e): Stability of $G(z) = 1/(1 - 1.5z^{-1})$

Pole at $z = 1.5$, which is *outside* the unit circle ($|1.5| > 1$).

If causal: ROC is $|z| > 1.5$. The unit circle ($|z| = 1$) is *not* in the ROC. System is **unstable**. Impulse response $g[n] = (1.5)^n u[n]$ grows without bound.

If anti-causal (left-sided): ROC is $|z| < 1.5$. The unit circle ($|z| = 1$) *is* in the ROC. System is **BIBO stable**. Impulse response $g[n] = -(1.5)^n u[-n-1]$ decays as $n \to -\infty$.

Key Point

This parallels the Laplace result from Problem 4(e) of the Lectures 16–18 homework: a pole outside the unit circle makes a causal system unstable but is compatible with a stable anti-causal system. The ROC determines stability, not the pole location alone.

Problem 5: The Unilateral z-Transform

Part (a): First-Order with IC

System: $y[n] - 0.8\,y[n-1] = (0.5)^n u[n], \quad y[-1] = 3.$

(i) Unilateral z-transform.

Left side: $Y(z) - 0.8\big[z^{-1}Y(z) + y[-1]\big] = Y(z) - 0.8z^{-1}Y(z) - 0.8(3) = (1 - 0.8z^{-1})Y(z) - 2.4$

Right side: $\frac{1}{1-0.5z^{-1}}$.

(ii) Solve for $Y(z)$:

$$(1 - 0.8z^{-1})Y(z) = \frac{1}{1 - 0.5z^{-1}} + 2.4$$

$$Y(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})} + \frac{2.4}{1 - 0.8z^{-1}}$$

(iii) Partial fractions on the first term:

$$\frac{1}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})} = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.8z^{-1}}$$

Set $z^{-1} = 2$: $1 = A(1 - 1.6) = -0.6A \Rightarrow A = -5/3$.

Set $z^{-1} = 1.25$: $1 = B(1 - 0.625) = 0.375B \Rightarrow B = 8/3$.

So:

$$Y(z) = \frac{-5/3}{1 - 0.5z^{-1}} + \frac{8/3}{1 - 0.8z^{-1}} + \frac{2.4}{1 - 0.8z^{-1}} = \frac{-5/3}{1 - 0.5z^{-1}} + \frac{8/3 + 12/5}{1 - 0.8z^{-1}}$$

$8/3 + 12/5 = 40/15 + 36/15 = 76/15$.

$$\boxed{y[n] = \left[-\frac{5}{3}(0.5)^n + \frac{76}{15}(0.8)^n\right] u[n]}$$

(iv) Verify:

$y[0] = -5/3 + 76/15 = -25/15 + 76/15 = 51/15 = 17/5 = 3.4$.

From ODE: $y[0] = 0.8 \cdot y[-1] + x[0] = 0.8(3) + 1 = 3.4$. ✓

$y[1] = -5/3(0.5) + 76/15(0.8) = -5/6 + 60.8/15 = -5/6 + 304/75$.

$= -62.5/75 + 304/75 = 241.5/75$... let us use decimals: $-0.833 + 4.053 = 3.22$.

From ODE: $y[1] = 0.8 \cdot y[0] + x[1] = 0.8(3.4) + 0.5 = 2.72 + 0.5 = 3.22$. ✓

Part (b): Second-Order Zero-Input

System: $y[n] - 0.5\,y[n-1] - 0.06\,y[n-2] = 0, \quad y[-1] = 5,\ y[-2] = 0.$

(i) Unilateral z-transform:

$$\mathcal{Z}_u\{y[n-1]\} = z^{-1}Y + y[-1] = z^{-1}Y + 5$$
$$\mathcal{Z}_u\{y[n-2]\} = z^{-2}Y + y[-2] + y[-1]z^{-1} = z^{-2}Y + 0 + 5z^{-1}$$

Substitute:

$$Y - 0.5(z^{-1}Y + 5) - 0.06(z^{-2}Y + 5z^{-1}) = 0$$
$$(1 - 0.5z^{-1} - 0.06z^{-2})Y = 2.5 + 0.3z^{-1}$$

Factor: $z^2 - 0.5z - 0.06 = 0 \Rightarrow z = \frac{0.5 \pm \sqrt{0.25+0.24}}{2} = \frac{0.5 \pm 0.7}{2}$.

Poles: $z = 0.6$ and $z = -0.1$.

$$Y(z) = \frac{2.5 + 0.3z^{-1}}{(1 - 0.6z^{-1})(1 + 0.1z^{-1})}$$

Partial fractions:

$$\frac{2.5 + 0.3z^{-1}}{(1 - 0.6z^{-1})(1 + 0.1z^{-1})} = \frac{A}{1 - 0.6z^{-1}} + \frac{B}{1 + 0.1z^{-1}}$$

Set $z^{-1} = 5/3$: $2.5 + 0.5 = A(1 + 1/6) = (7/6)A \Rightarrow 3 = (7/6)A \Rightarrow A = 18/7$.

Set $z^{-1} = -10$: $2.5 - 3 = B(1 + 6) = 7B \Rightarrow -0.5 = 7B \Rightarrow B = -1/14$.

$$\boxed{y[n] = \left[\frac{18}{7}(0.6)^n - \frac{1}{14}(-0.1)^n\right] u[n]}$$

Verify: $y[0] = 18/7 - 1/14 = 36/14 - 1/14 = 35/14 = 2.5$.

From ODE: $y[0] = 0.5 \cdot y[-1] + 0.06 \cdot y[-2] = 0.5(5) + 0 = 2.5$. ✓

(iii) The input is zero, so this is a **pure zero-input response**: the output is entirely due to the initial conditions $y[-1] = 5$ and $y[-2] = 0$.

Part (c): ZSR/ZIR Decomposition for Part (a)

From part (a): $Y(z) = \frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})} + \frac{2.4}{1-0.8z^{-1}}$.

Zero-state response (set $y[-1] = 0$, keep input):

$$Y_{\text{ZS}}(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})}$$

From the partial fractions computed in part (a): $A = -5/3$, $B = 8/3$.

$$\boxed{y_{\text{ZS}}[n] = \left[-\frac{5}{3}(0.5)^n + \frac{8}{3}(0.8)^n\right] u[n]}$$

Zero-input response (set input to 0, keep $y[-1] = 3$):

$$(1 - 0.8z^{-1})Y_{\text{ZI}} = 2.4 \quad \Rightarrow \quad Y_{\text{ZI}}(z) = \frac{2.4}{1 - 0.8z^{-1}}$$

$$\boxed{y_{\text{ZI}}[n] = 2.4\,(0.8)^n\,u[n]}$$

Verify the sum:

$$y_{\text{ZS}} + y_{\text{ZI}} = \left[-\frac{5}{3}(0.5)^n + \frac{8}{3}(0.8)^n + 2.4(0.8)^n\right] u[n] = \left[-\frac{5}{3}(0.5)^n + \left(\frac{8}{3} + \frac{12}{5}\right)(0.8)^n\right] u[n]$$

$\frac{8}{3} + \frac{12}{5} = \frac{40+36}{15} = \frac{76}{15}$.

This matches the total $y[n]$ from part (a). ✓

Key Point

The ZSR starts at $y_{\text{ZS}}[0] = -5/3 + 8/3 = 1 = x[0]$ (as if the system started at rest). The ZIR starts at $y_{\text{ZI}}[0] = 2.4$ (the effect of the IC). Their sum gives $y[0] = 1 + 2.4 = 3.4$, matching the ODE check.

Rogelio Gracia Otalvaro